Rohit Sachdeva

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Education

Master of Computer Applications (MCA) Sept 2025 – Present
G.L. Bajaj Institute of Technology and Management (AKTU) *Greater Noida, India*

Bachelor of Computer Applications (BCA) Sept 2021 – Aug 2024
ITS Mohan Nagar College (CCS University) *Ghaziabad, India*
CGPA: 7.2

Technical Skills

Languages: Python, Java, C++, SQL **Cloud**: Azure, Databricks
AI & Deep Learning: Neural Networks, Machine Learning
Computer Vision: OpenCV, Haar Cascade Classifiers, Image Pre-processing, Pattern Recognition
Tools: Git/GitHub, Jupyter Notebook, VS Code, Power BI, PyCharm

Projects

Immutable Scan: Medical Image Verification Framework | ***TensorFlow, OpenCV, Python*** Deep Learning, Cryptograph
Research Project (Deep Learning & Cryptography)

- Designed a novel **Neural Semantic Hashing** pipeline using a MobileNetV2 backbone to secure distributed Internet of Medical Things (IoMT) networks against adversarial manipulation.
- Engineered an **Integrity Guard** defense mechanism using deep feature hashing and Gaussian filters, achieving a **100% protection rate** on ultrasound data (BreastMNIST).
- Developed a **Dynamic Calibration** protocol that successfully restored adversarial detection rates from 0% to >**90% accuracy** on high-resolution radiography (PneumoniaMNIST).
- Formulated a **Label-Blind Entropy-Maximization Attack** model to evaluate system robustness against uninformed adversaries, benchmarking a performance boundary of ≤**74% accuracy** on dense skin cell histology (HAM10000).

Behavior-Based Early Career Prediction System | ***Python, Scikit-learn, TensorFlow, Databricks, Flask, SQLite***

- Engineered a hybrid-cloud pipeline using **PySpark** to ingest O*NET labor data into a **Databricks Gold Layer**, syncing records locally via the **Databricks SQL Connector** for training.
- Resolved the algorithmic **"Personality-Logic Gap"** by deploying a bimodal ensemble engine (**XGBoost + Deep Neural Network**) trained on scaled, 8-dimensional feature vectors (Math, Programming, and 60-question RIASEC traits).
- Developed an economic future-proofing mechanism using **Facebook Prophet** to execute **Time-Series Forecasting** on historical hiring volumes, outputting a custom 2030 career **Viability Index**.
- Built a secure web deployment using **Flask** and an optimized **SQLite** authentication layer handling hashed user sessions and real-time state persistence across multi-stage assessment dashboards.

Face Detection System with Audio Feedback Python, OpenCV, gTTS
Computer Vision Project

- Implemented a real-time detection pipeline using **Haar Cascade Classifiers** to identify and count frontal faces in imagery.
- Integrated **Google Text-to-Speech (gTTS)** to bridge visual processing with audio feedback for a multimodal user experience.

Experience

Plan India Feb 2025 – July 2025
Data Analyst Intern *New Delhi, India*

- **Automation Engineering:** Built a custom Macro-based script to convert complex Word document templates into standardised Excel datasets.
- **System Optimisation:** Optimised the internal Knowledge Repository (SharePoint) with advanced search logic and structured metadata filters.
- **Technical Research:** Conducted feasibility analysis for integrating AI-based chatbots (Azure Bot Service & Power Virtual Agents) to automate information retrieval.
- Developed analytical dashboards using **Power BI** to visualise organisational KPIs and project impact.

Click at Career June 2024 – Jan 2025
Student Mentor (Technical) *Remote*

- Mentored undergraduate students in Python programming and Data Analytics fundamentals.
- Debugged code and guided students through algorithm implementation challenges.

Interests

Mentoring, Big Data, Machine Learning